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Computer Architecture Assessment - 1

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①

Given the following register and memory contents.

Register	value (Hex)
PC	4200H
R1	1500H
R2	2500H
IX	0800H
SP	3000H

memory contents

Address Content

1500H	45H
1502H	3200H
2500H	18H
2800H	3400H
3000H	3600H
3200H	55H
3400H	66H
3600H	77H
4204H	200H

Questions :-

Determine the effective Address (EA) and operand value usage.

- Immediate
- Direct
- Indirect
- Register
- Register indirect
- Indexed
- Relative
- stack Addressing

Assume

Addressing mode	Effective Address	Operand Value	Explanation
Immediate	NO EA	200H	
Direct	0200H	NOT SPECIFIED	
Indirect	M[0200H]	CANNOT BE DETERMINED	
Register (Ri)	R1	1500H	
Register indirect	1500H	45H	
Indexed (1x200H)	0800H + 0200H = 0A00H	0A00H memory not given	
Relative (PC + 200H)	4200H + 0200H = 4400H	4400H memory not given.	
State Address	3000H	3000H	

2. Number of instruction (IC) = 900 million = 900×10^6
Average CPI = 2.8

Initial clock Rate = 2.4 GHz = 2.4×10^9 Hz

upgraded clock Rate = 3.6 GHz = 3.6×10^9 Hz

CPI remains 2.8

$$\text{Execution time} = \frac{\text{Instruction Count}}{\text{Clock Rate}} \times \text{CPI}$$

a) original execution time

$$= \frac{900 \times 10^6 \times 2.8}{2.4 \times 10^9} = \frac{900 \times 2.8}{2.4 \times 10^3} = \frac{9 \times 10^2 \times 2.8}{2.4 \times 10^3}$$

$$= \frac{25.2}{2.4 \times 10} = \frac{2.52}{2.4} = 1.05 \text{ s}$$

b) new execution time

$$= \frac{900 \times 10^6 \times 2.8}{3.6 \times 10^9} = \frac{900 \times 2.8}{3.6 \times 10^3} = \frac{9 \times 2.8}{3.6 \times 10^1}$$

$$= \frac{25.2}{3.6 \times 10} = \frac{2.52}{3.6} = 0.7 \text{ s}$$

c) Speed up

$$\text{Speed up} = \frac{\text{old execution time}}{\text{new execution time}} = \frac{1.05}{0.70} = 1.5$$

$$= 1.5 \times (50\% \text{ faster})$$

d) Revised execution time (CPI increase by 10%)

$$\text{new CPI} = 2.8 \times 1.10 = 3.08$$

$$\text{Execution time} = \frac{900 \times 10^6 \times 3.08}{3.6 \times 10^9} = \frac{2772}{3606} = 0.771$$

e) is increasing clock frequency always:-
 although increasing the clock frequency reduced time from 1.05s to 0.70s if highest frequency cause the CPI to increase 10%.

3) A processor executes a program with instruction mix

instruction type	instruction count	CPI
arithmetic	5×10^6	2
memory access	3×10^6	5
Branch	1×10^6	4
floating point	2×10^6	8

- Calculate the total no. of clock cycle Required
- Calculate the CPU execution time
- Determine the average CPI

- d) if the memory access CPI is reduced form
 e) calculate the percentage improvement

Sol:

a) Total clock cycle = $\sum C \times CPI$

$$= (5 \times 10^6 \times 2) + (1 \times 10^6 \times 4) + (2 \times 10^6 \times 8) + (5 \times 10^6 \times 5)$$

$$= (10 \times 10^6) + (4 \times 10^6) + (16 \times 10^6) + (25 \times 10^6)$$

$$= 10^6 (10 + 4 + 16 + 25)$$

$$= 45 \times 10^6 \text{ cycles}$$

b) execution time = $\frac{\text{Total clock cycle}}{\text{clock rate}}$

$$= \frac{45 \times 10^6}{2.5 \times 10^7}$$

$$= 0.018 \text{ sec}$$

$$= 0.018 \text{ sec}$$

c) average CPI = $\frac{\text{Total clock cycle}}{\sum C}$

$$= \frac{45 \times 10^6}{(5 + 3 + 2) \times 10^6} \Rightarrow \frac{45 \times 10^6}{10 \times 10^6}$$

$$= 4.5$$

$$CPI = 4.09$$

d) clock cycle = $(5 \times 10^6 \times 2) + (3 \times 10^6 \times 3) + (1 \times 10^6 \times 4) + (2 \times 10^6 \times 8)$

$$= (10 + 9 + 4 + 16) \times 10^6$$

$$= 39 \times 10^6$$

execution time = $\frac{3.9 \times 10^6}{2.5 \times 10^7}$

$$= 0.0156 \text{ sec}$$

④

Processor A:

- clock frequency = 4 GHz
- CPI = 2.5
- Instruction count = 500 million

Processor B:

- clock frequency = 3 GHz
- CPI = 1.8
- Instruction count = 550 million

Processor C:

- clock frequency = 5 GHz
- CPI = 3.1
- Instruction count = 420 million

Questions:-

1. Calculate the CPU execution time of processor
2. Rank the processor based on execution time
3. compute the speed up
4. If processor B reduces its instruction count by 10%, determine its new execution time
5. Recommend the best processor, justifying your decision based on clock frequency memory trace problem

Sol:-

$$\text{CPU execution time} = \frac{\text{Instruction count} \times \text{CPI}}{\text{clock frequency}}$$

1. Processor A:

$$\frac{500 \times 10^6 \times 2.5}{4 \times 10^9} = 0.3125 \text{ sec}$$

Processor B:

$$\frac{550 \times 10^6 \times 1.8}{3 \times 10^9} = 0.33 \text{ sec}$$

Ranking: fastest $\rightarrow$ slowest

3. speed up of fastest over slowest

Fastest = c, slowest = B

$$\text{speed up} = \frac{0.33}{0.2666}$$

speed up ≈ 1.24 , approx

4. processor B reduces instruction count by 10%

new instruction count:

$$550 \times 0.90 = 495 \text{ million}$$

new execution time:

$$\frac{495 \times 10^6 \times 1.8}{3 \times 10^9} = 0.297 \text{ sec}$$

5. Recommendation

Processor c is best choice because it has the lowest execution time (0.2666 s), despite the lowest having higher CPI. it also has the highest frequency and relatively low instruction count

(5)

memory contents before execution!

Address Data

2500 H

15 H

2501H 25H

Program

LDA 2500H

MOV B, A

LDA 2501H

ADD B

STA 2502H

HLT

Questions :-

1. Determine the contents of A and B after each instruction
2. What value is stored in 2502H?
3. Which flags are modified?
4. Draw the execution trace.

Sol :-

1) contents of A and B after each instruction

Instruction	A	B
LDA 2500H	15H	15H
MOV B, A	15H	15H
LDA 2501H	25H	15H
ADD B	3AH	15H
STA 2502H	3AH	15H
HLT	3AH	15H

For ADD B:

$$25H + 15H = 3AH$$

2) value stored in 2502H

$$2502H = 3AH$$

3) which flags are modified

The ADD B instruction modifies

- Sign (S)
- Zero (Z)
- Auxiliary Carry (AC)
- Parity (P)
- Carry (CY)

$$\text{For } 25H + 15H = 3AH:$$

- S = 0
- Z = 0
- AC = 0
- PF = 1
- CY = 0

4) plain text

$$2500H = 15H$$

↓

LDA 2500H

↓

$$A = 15H$$

↓

MOV B, A

↓

$$B = 15H$$

↓

ADD B

↓

$$A = 25H + 15H = 3AH$$

↓

STA 2502H

$$2502H = 3AH$$

↓

HLT

Final Answer:

$$A = 3AH$$

$$B = 15H$$

$$2502H = 3AH$$